ANALOGUE ELECTRONICS IV

EEE 434

LECTURE NOTES ON

(Principles of operation, classification, construction and characteristics of oscillators).

2005

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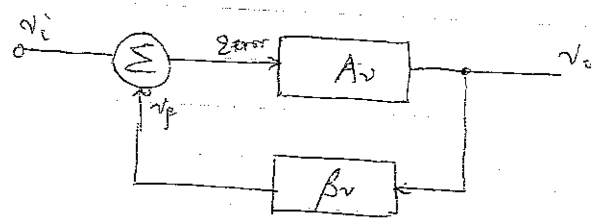
Positive feedback Concept

One of the characteristics of an oscillator is that it must have a positive feedback which leads to high gain, a possibility of oscillation. Below is a block diagram of an amplifier with positive feedback



It has been shown that

$$A_{vf} = \frac{A_v}{1 - A_v\beta_v} \quad \cdots\cdots \text{eqn } ①$$

Criteria for oscillations

1) The Barkhausen Criterion must be satisfied, that is

$$A\beta = 1$$

If $A\beta < 1$, oscillation will not or will cease to occur.
If $A\beta \gg 1$, a non-pure sine wave is not achieved.

2) The frequency at which a sinusoidal oscillator will operate is the frequency for which the total phase shift introduced, as a signal proceeds from the input terminals, through the amplifier and feedback network, and back again to the input, is precisely zero (or integer multiple of 2π). OR simply, the frequency of a sinusoidal oscillator is determined by the condition that the loop phase shift is zero.

From eqn 1, when $A\beta = 1$

$$A_{vf} = \frac{A_v}{1-1} = \frac{A_v}{0} = \infty$$

$$\text{Since } A_{vf} = \frac{A_v}{0} = \frac{V_o}{V_i},$$

It can be analysed to mean that there will be an output even without input (ie $V_i = 0$). It has become an

Oscillator (signal generator).

Oscillators (Introduction)

Many electronic devices require a source of energy at a specific frequency from few Hz to several MHz. This is only achieved by a device called oscillator. The output waveform of oscillator may be sinusoidal, triangular (saw tooth), pulse and square wave depending on application. They can be used to generate high frequency as carrier waves in Radio and TV applications, radar electronic computers and devices to mention but few.

Sinusoidal oscillators

A transistor can work as an oscillator to produce continuous undamped oscillations of any desired frequency if frequency-determining and feedback circuits are properly connected to it. The major difference between these oscillators lie in

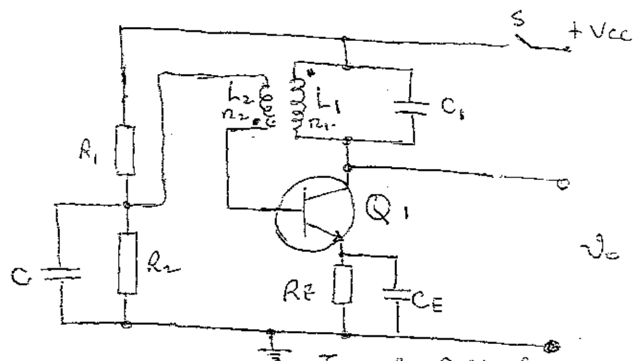
i) the frequency determining elements (L-C or R-C combination)
ii) method of providing positive feedback.

L-C tank oscillators or Tuned oscillators

These can be either Tuned Collector, tuned base, Hartley Colpitts or Clapp.

Tuned Collector oscillator

The diagram below shows the circuit of tuned collector oscillator. It contains tuned $L_1 - C_1$ in the collector and hence the name. The feedback coil L_2 in the base circuit is magnetically coupled to the tank circuit coil L_1. In practice, L_1 and L_2 form the primary and secondary of the transformer. The biasing is provided by potential divider arrangement. The capacitor C connected in the base circuit provides low reactance path to the oscillators



Tuned Collector oscillator

Circuit operation

When switch S is closed, the collector current starts increasing and charges the capacitor C_1. When fully charged it discharges through coil L_1, setting up oscillations of frequency given by

$$f_o = \frac{1}{2\pi\sqrt{L_1C_1}}$$

These oscillations induce some voltage in the coil L_2 by mutual induction. The frequency of voltage in the coil L_2 is the same as that of tank circuit but its magnitude depends upon the number of turns of L_2 and coupling between L_2 and L_1. The voltage across L_2 is applied between base and emitter and appears in the amplified form in the collector circuit, thus overcoming the losses occurring in the tank circuit. The number of turns of L_2 and coupling between L_1 and L_2 are so adjusted that oscillations across L_2 are amplified to a level just sufficient to supply losses to the tank circuit. It is worth mentioning that energy supplied to the tank circuit is in phase with generated oscillations since a phase shift of 180° is created between voltages L_1 and L_2. A further phase shift of 180° takes place between base-emitter and collector circuit due to transistor properties

Frequency of operation

At resonance frequency f_o, it is known that the inductive reactance = capacitive reactance ie

$$X_L = X_C$$

(Parallel resonance is $Z_o = L/CR$ $\therefore V^{\uparrow}$)

$$2\pi f_o L_1 = \frac{1}{2\pi f_o C_1}$$

$$4\pi^2 f_o^2 L_1 C_1 = 1$$

$$f_o^2 = \frac{1}{4\pi^2 L_1 C_1} \quad \therefore \quad f_o(Hz) = \frac{1}{\sqrt{4\pi^2 L_1 C_1}} = \frac{1}{2\pi\sqrt{L_1C_1}}$$

The feedback factor, β_v is

$$\beta_v = \frac{n_2}{n_1}$$

for Barkhausen Criterion,

$$A_v \beta_v = 1$$

i.e.

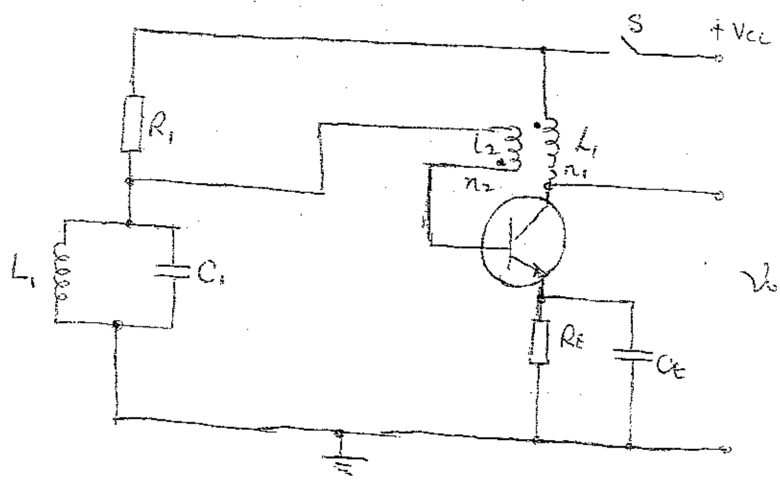
$$\frac{n_2}{n_1} A_v = 1$$

$$A_v = \frac{n_1}{n_2}$$

$\therefore A_v \geq \frac{n_1}{n_2}$ for oscillation to start.

Tuned Base Oscillator

This is given below:



Tune base oscillator

At resonance frequency f

$$\omega_o L_1 = \frac{1}{\omega_o C_1}$$

$$\omega_o^2 L_1 C_1 = 1$$

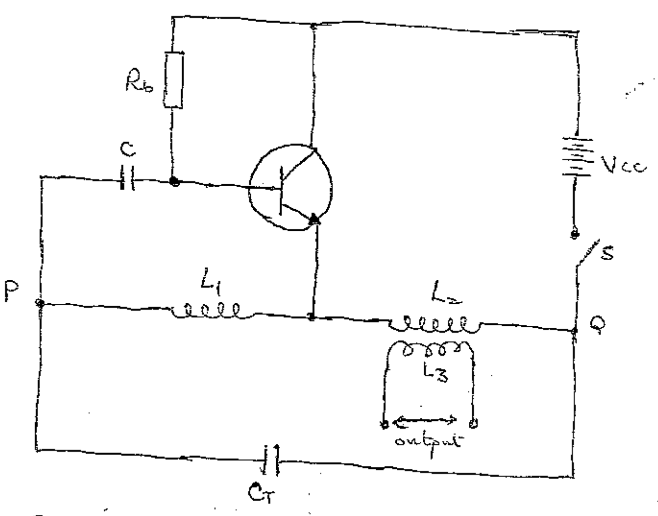
$$\omega_o^2 = \frac{1}{L_1 C_1}$$

$$(2\pi f_o)^2 = \frac{1}{L_1 C_1}$$

$$\therefore f_o = \frac{1}{2\pi\sqrt{L_1 C_1}}$$

Hartley oscillator

This oscillator is very popular in applications where substantial radio frequency power is needed as in radio transmitters, induction heating and remote control systems. Below is the diagram of Hartley oscillator.



The tank circuit is made up of L_1, L_2 and ~~of~~ C_T. The coil L_1 is inductively coupled to coil L_2 the combination functions as an auto-transformer. The resistance R_b between collector and base provides the necessary biasing. The capacitor C blocks the dc component. The frequency of oscillation is given by

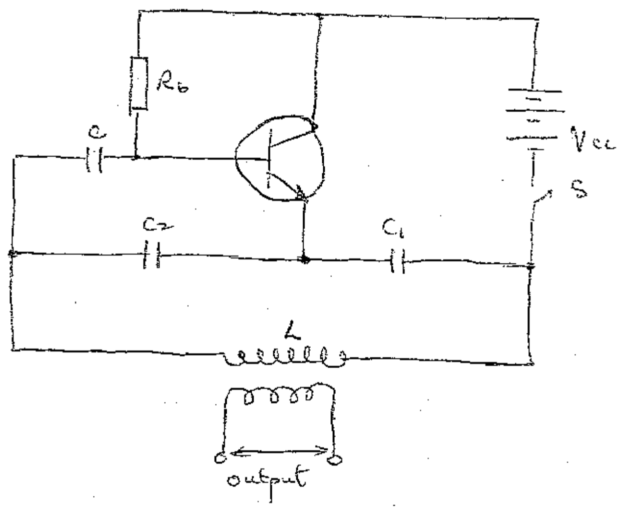
$$f_o = \frac{1}{2\pi\sqrt{C_T(L_1+L_2)}}$$

Circuit operation.

When switch S is closed, the collector current starts rising and charges the capacitor C_T. When it is fully charged, it discharges through coils L_1 and L_2, setting up oscillations of frequency determined by the above formula. The oscillations across L_1 are applied to the base emitter junction and appear in the amplified form in the collector circuit. The coil L_2 couples the collector circuit energy back into the tank by means of mutual inductance between L_1 and L_2. In this way energy is being continuously supplied to the tank circuit to overcome the losses occuring in it. It may be seen that the energy supplied to the tank circuit is of correct phase. The ends P and Q of the auto-transformer $L_1 - L_2$ are 180° out of phase. A further phase-shift of 180° is produced by base and collector circuit of the transistor. In this way, energy fed back to the tank circuit is in phase with the generated oscillations.

Collpitt's Oscillator

The Collpitts oscillator is similar to Hartley oscillator with minor modifications. Whereas Hartley oscillator uses two coils and a capacitor for frequency selection, Collpitts oscillator uses two ~~coils~~ capacitors and a coil as shown below:



The tank circuit is made up of C_1, C_2 and L. The frequency of the oscillations is determined by the values of C_1, C_2 and L and is given by

$$f = \frac{1}{2\pi\sqrt{L C_T}}$$

$$\text{where } C_T = \frac{C_1 C_2}{C_1 + C_2}$$

Circuit operation: When switch S is closed, the capacitors C_1 and C_2 are charged. These capacitors are discharged through coil L, setting up oscillations determined by the above equation. The oscillations across C_2 are applied to the base-emitter junction and will appear in the amplified form in the collector circuit and supply losses in the tank circuit. The amount of feedback depends upon the relative

Hartley oscillator

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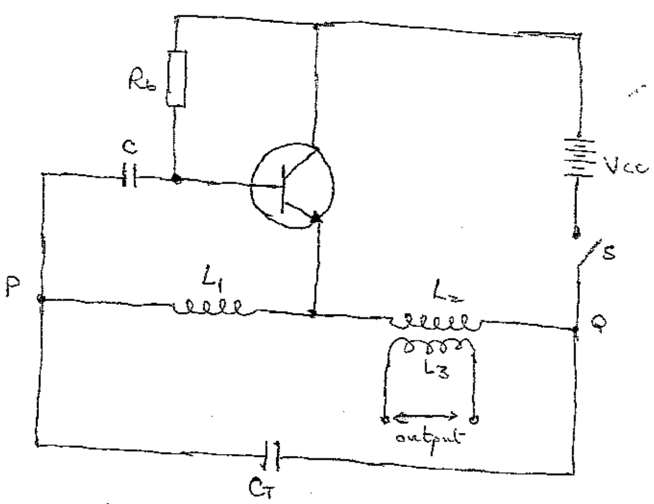


The tank circuit is made up of L_1, L_2 and ~~C~~ C_T. The coil L_1 is inductively coupled to coil L_2 the combination functions as an auto-transformer. The resistance R_b between collector and base provides the necessary biasing. The capacitor C blocks the dc component. The frequency of oscillation is given by

$$f_0 = \frac{1}{2\pi\sqrt{C_T(L_1+L_2)}}$$

Circuit operation.

When switch S is closed, the collector current starts rising and charges the capacitor C_T. When it is fully charged, it discharges through coils L_1 and L_2, setting up oscillations of frequency determined by the above formula. The oscillations across L_1 are applied to the base emitter junction and appear in the amplified form in the collector circuit. The coil L_2 couples the collector circuit energy back into the tank by means of mutual inductance between L_1 and L_2. In this way energy is being continuously supplied to the tank circuit to overcome the losses occurring in it. It may be seen that the energy capacitance values C_1 and C_2. The smaller the capacitance C_1, the greater the feedback. The capacitors C_1 and C_2 act as a simple alternating voltage divider. Therefore points P and Q are 180° out of phase. A further phase shift of 180° is produced by the transistor. In this way, feedback is properly phased to produce continuous undamped oscillations.

R-C phase shift Oscillator

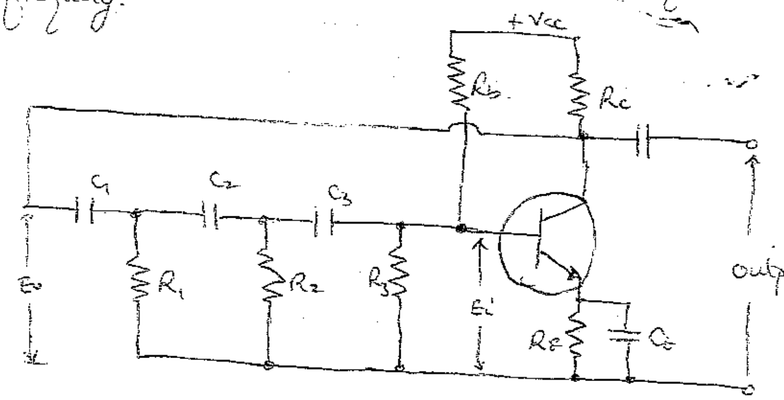
These oscillators are used as signal generators from lowest audio range up to around 10 MHz. Below is a circuit of phase shift oscillator. It consists of a conventional single transistor amplifier and a RC phase shift network. The phase shift network consists of three sections R_1C_1, R_2C_2, and R_3C_3. At some frequency f_1 the phase shift produced by each section is 60° so that total phase shift produced by the RC network is 180°. The frequency of oscillations is given by

$$f_0 = \frac{1}{2\pi RC\sqrt{6}}$$

where $R_1 = R_2 = R_3 = R$

$C_1 = C_2 = C_3 = C$

Because the capacitive reactance of the network change at different frequencies, 180° phase shift required occurs only at one frequency.



Circuit operation: When the circuit is switched on, it produces oscillations of frequency determined by the formula given above. The output E_o of the amplifier is feedback to RC feedback network. This network produces a phase shift of 180° and a voltage E_i appears at its output which is applied to the transistor amplifier with correct phase. A phase shift of 180° is produced by the transistor amplifier. A further phase shift of 180° is produced by the RC network. As a result, the phase shift around the entire loop is 360° i.e. feedback is positive.

Advantages

- It does not require transformers or inductors
- It can be used to produce very low frequencies
- The circuit provides good frequency stability.

Disadvantages

- It is difficult for the ckt to start oscillations
- The circuit gives small output.

Circuit Analysis

It can be seen that the voltage gain of the amplifier

$$A_v = \frac{v_o}{v_i} = -\left(\frac{5}{\omega^2R^2C^2} - 1\right) - j\left(\frac{6}{\omega RC} - \frac{1}{\omega^3R^3C^3}\right) \quad \text{--- (1)}$$

Since the second term, the reactive j term, must be zero at frequency of oscillation, f_o, the gain expression V_o/v_i becomes

$$\frac{V_o}{v_i} = -\left(\frac{5}{\omega^2R^2C^2} - 1\right) \quad \text{- - - - - (2)}$$

The second term in eqn (1) can be used to solve for f_o,

i.e. $$\frac{6}{\omega_o RC} - \frac{1}{\omega_o^3R^3C^3} = 0 \quad \text{- - - (3)}$$

$$\frac{6}{\omega_o RC} = \frac{1}{\omega_o^3R^3C^3}$$

$$\omega_o^3R^3C^3\,6 = \omega_o RC$$

$$\omega_0^2 = \frac{RC}{R^3C^3 \times 6}$$

$$\omega_0^2 = \frac{1}{6R^2C^2}$$

$$\omega_0 = \frac{1}{\sqrt{6}\,RC} \quad \cdots\cdots (4)$$

$$\therefore f_0 = \frac{1}{2\pi\sqrt{6}\,RC} \text{ Hertz} \quad \cdots\cdots (5)$$

At oscillation frequency

$$\frac{v_o}{v_i} = -\left(\frac{5}{\omega_0^2 R^2C^2} - 1\right)$$

Substituting for ω_0

$$= -\left(\frac{5}{(1/6R^2C^2)R^2C^2} - 1\right)$$

$$= -\left(\frac{5(6R^2C^2)}{R^2C^2} - 1\right)$$

$$= -(30 - 1)$$

$$= -29$$

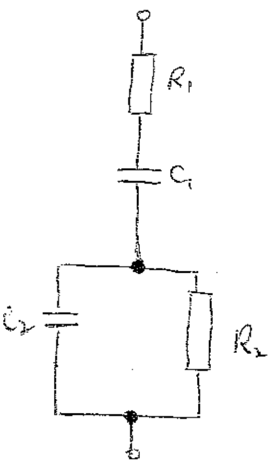
Since $A_v\beta_v = 1$

$\beta_v = -\frac{1}{29}$ $\therefore$ for sustainable oscillations $A_v = -29$

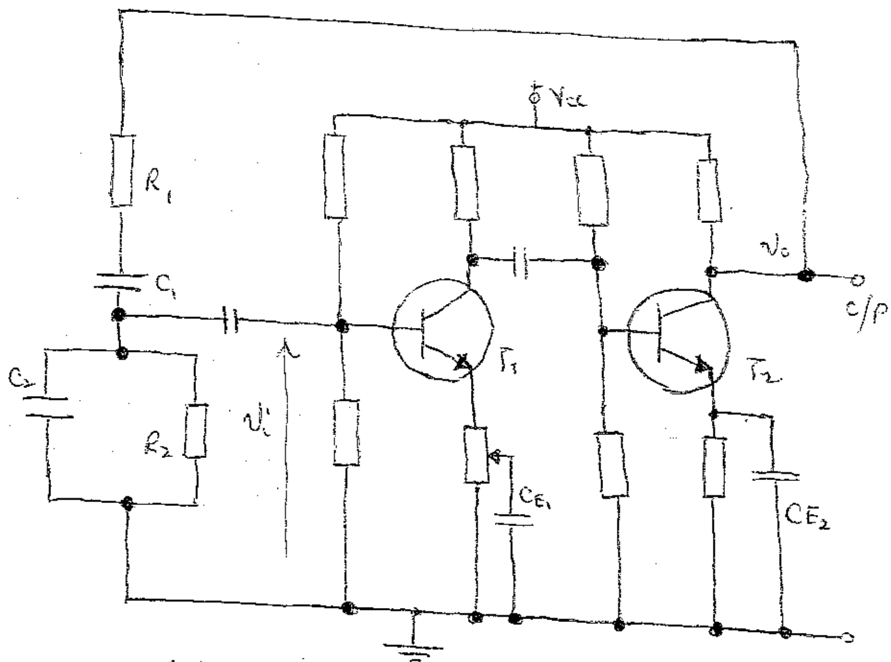
or slightly higher. i.e. $A_v \geq -29$

WIEN BRIDGE OSCILLATOR

This is a standard oscillator circuit for all low frequencies in the range of 10 Hz to about 1 MHz. It's the most frequently used type of audio oscillator as the output is free from circuit fluctuations and ambient temperature. The circuit is given as below:-



a) Wien bridge



(b) Wien-bridge oscillator

Like phase-shift, the RC circuit is used as the frequency-selective network; however, the RC Wien bridge's operation is quite different. The Wien-bridge oscillator uses either a two-stage amplifier or an operational amplifier. T_1 serves as an oscillator and amplifier while the other transistor T_2 serves as an inverter (i.e. produces a phase shift of 180°). The circuit uses positive and negative feedbacks. The positive feedback is through R_1C_1, R_2C_2 to the transistor T_1. The negative feedback is through the voltage dividers to the input of transistors T_1 and T_2 respectively to ensure constant output. At the frequency of operation f_0, the reactive branch produces a phase lead and phase lag voltage cancellation. Now if $R_1 = R_2 = R$ and $C_1 = C_2 = C$ the bridge provides a voltage divider, and so, the fraction output voltage feedback through the divider to the first transistor T_1 is given by

$$\beta_v = \frac{R\left(-j\frac{1}{\omega C}\right)/\left(R-j\,1/\omega C\right)}{\left(R-j\,1/\omega C\right)+\left[R\left(-j\,1/\omega C\right)/\left(R-j\frac{1}{\omega C}\right)\right]} \quad \cdots (1)$$

Since $\beta_v = \frac{v_i}{v_o}$

Equation (1) simplifies to

$$\frac{R\left(-j\frac{1}{\omega C}\right)/\left(R-j\frac{1}{\omega C}\right)}{\left\{\left(R-j\frac{1}{\omega C}\right)+\left[R\left(-j\frac{1}{\omega C}\right)/\left(R-j\frac{1}{\omega C}\right)\right]\right\}} \times \frac{\left(R-j\frac{1}{\omega C}\right)}{\left(R-j\frac{1}{\omega C}\right)}$$

$$= \frac{R\left(-j\frac{1}{\omega C}\right)}{\left(R-j\frac{1}{\omega C}\right)^2 - j\frac{R}{\omega C}}$$

$$= \frac{-j\frac{R}{\omega C}}{R^2-2j\frac{R}{\omega C}+j^2\frac{1}{\omega^2C^2} - j\frac{R}{\omega C}}$$

$$= \frac{-j\,R/\omega C}{R^2-3j\frac{R}{\omega C}-\frac{1}{\omega^2C^2}} \times \frac{j}{j}$$

$$= \frac{-j^2 R/\omega C}{jR^2-3j^2R/\omega C - j\frac{1}{\omega^2C^2}}$$

$$= \frac{R/\omega C}{3R/\omega C + j\left(R^2 - 1/\omega^2C^2\right)} \quad \cdots (2)$$

At frequency of operation, f_o, the j term becomes zero ie no reactive part but only real part ie $(R^2 - 1/\omega_o^2 C^2) = 0$

$$\text{ie } \beta_v = \frac{R/\omega_o C}{3R/\omega_o C} = 1/3 \quad \cdots ③$$

Now solving for f_o using reactive j term of eqn ② yields

$$R^2 - \frac{1}{\omega_o^2 C^2} = 0$$

$$R^2 = \frac{1}{\omega_o^2 C^2}$$

Hence,

$$\omega_o^2 = \frac{1}{R^2 C^2}$$

$$\therefore f_o = \frac{1}{2\pi RC} \quad \text{Hertz.}$$

It can be deduced from eqn ③ that for the magnitude of A_v for sustainable oscillation is given from

$$A_v \beta_v = 1$$

$$A_v \cdot 1/3 = 1$$

$$\therefore A_v \geq 3$$

The two transistors T_1 and T_2 must give overall gain of at least 3. Since T_2 is just to give only phase inversion, it can be arranged to give amplification of 1 ie (Unity gain) and T_1 can be arranged to give amplification of 3 or more (overall gain $G_T = G_1 \times G_2 = 3 \times 1 = 3$) hence the need for gain adjustment on T_1 and feedbacks on both T_1 and T_2.

<u>Advantages</u>: It gives constant output, the circuit works quite easily, the overall gain is high because of two transistors and the frequency of oscillations can be changed easily by using a potentiometer.

<u>Disadvantages</u>: The circuit requires 2 transistors and large number of components and it cannot generate very high frequencies (VHF)

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Limitations of LC and RC oscillators

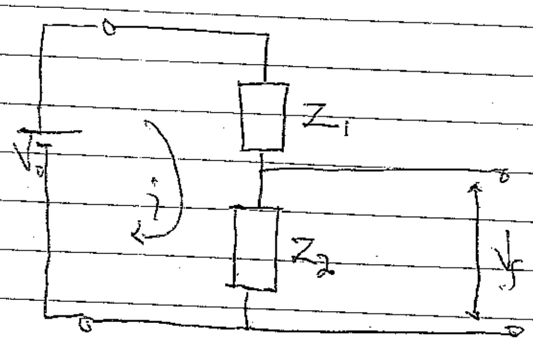
The LC and RC discussed earlier have their own limitations. The major problem in such circuit is that their frequency of operation does not remain strictly constant. It's because the values of resistors and inductors which are the frequency determining elements in these circuits, change with temperature. This cause the change in the frequency of the oscillator. However in many applications it's desirable and necessary to maintain the frequency constant with extreme low tolerances. For example, it is desired that frequency of the broadcasting station should not change due to change in temperature or any other reason or else listeners will ever be tuning their radios for any particular station.

Solution

In order to maintain constant frequency, piezoelectric crystals are used in place of LC or RC circuits as temperature effect on the crystals is very minimal. Oscillators of this type are called **CRYSTAL OSCILLATORS**. For less serious work, LC and RC can be used with (i) provide frequency-determining network, with low tolerance components (ii) provide feedbacks to combat the effect of temperature.

TUTORIALS

$V_i = V_f \Rightarrow$ WIEN BRIDGE OSCILLATOR



$V_f = V_i \quad , \quad V_f = i \cdot Z_2$

$$i = \frac{V_0}{Z_1 + Z_2} \quad \therefore \quad V_f = \frac{V_0 \cdot Z_2}{Z_1 + Z_2}$$

$$\beta = \frac{V_f}{V_0} = \frac{Z_2}{Z_1 + Z_2}$$

$$Z_1 = R + \frac{1}{j\omega C}$$

$$Z_1 = \frac{j\omega RC + 1}{j\omega C}$$

$$Z_1 = \frac{1 + j\omega RC}{j\omega C}$$

$$Z_2 = \frac{R \cdot \frac{1}{j\omega C}}{R + \frac{1}{j\omega C}}$$

$$Z_2 = \frac{R/j\omega C}{\frac{1 + j\omega RC}{j\omega C}}$$

$$Z_2 = \frac{R}{\cancel{j\omega C}} \times \frac{\cancel{j\omega C}}{1 + j\omega RC}$$

$$Z_2 = \frac{\cancel{j\omega RC}}{\cancel{j\omega C + j^2\omega^2 C^2 R}} = \frac{R}{1 + j\omega RC}$$

$$Z_2 = \frac{\cancel{j\omega RC}}{\cancel{j\omega C - \omega^2 C^2 R}}$$

$$\frac{V_f}{V_0} = \frac{\frac{R}{1 + j\omega RC}}{\frac{1 + j\omega RC}{j\omega C} + \frac{R}{\cancel{1 + j\omega RC}}} \times \frac{\cancel{1 + j\omega RC}}{\cancel{1 + j\omega RC}}$$

$$= \frac{R}{\frac{(1 + j\omega RC)^2}{j\omega C} + R}$$

$$= \frac{R}{\frac{(1 + j\omega RC)^2 + j\omega RC}{j\omega C}}$$

1

$$= \frac{jKwC}{1 + jwRC + jwRC + j^2 w^2 R^2 C^2 + jwRC} = \frac{jwRC}{1 + 3jwRC - w^2 R^2 C^2}$$

$$\frac{V_f}{V_0} = \frac{jwRC}{1 - w^2 R^2 C^2 + 3jwRC} \times \frac{j}{j}$$

$$= \frac{-wRC}{j - jw^2 R^2 C^2 - 3wRC}$$

$$= \frac{wRC}{-j + jw^2 R^2 C^2 + 3wRC}$$

$$\frac{V_f}{V_0} = \frac{wRC}{3wRC - j(1 - w^2 R^2 C^2)}$$

$$1 - w^2 R^2 C^2 = 0$$

$$w^2 R^2 C^2 = 1$$

$$w^2 = \frac{1}{R^2 C^2}$$

$$w = \frac{1}{RC}$$

$$f = \frac{1}{2\pi RC}$$

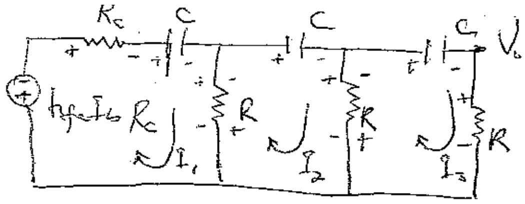
$$\frac{V_f}{V_0} = \beta = \frac{w\cancel{RC}}{3w\cancel{RC}} = \frac{1}{3}$$

$$A\beta = 1$$

$$\frac{1}{3}A = 1$$

$$A \geq 3$$

2



KVL to loop ①, ② & ③

Loop ①

$-h_{fe}I_bR_c - I_1R_c - I_1\cdot\frac{1}{j\omega c} - I_1R + I_2R = 0$

$h_{fe}I_bR_c + I_1R_c + I_1\frac{1}{j\omega c} + I_1R - I_2R = 0$

$I_1\left[R+R_c+\frac{1}{j\omega c}\right] - I_2R + h_{fe}I_bR_c = 0$

but $I_b = I_3$

$I_1\left[R+R_c+\frac{1}{j\omega c}\right] - I_2R + I_3h_{fe}R_c = 0$ ← ①

Loop ②

$-I_2\cdot\frac{1}{j\omega c} - I_2R + I_3R - I_2R + I_1R = 0$

$-I_2\frac{1}{j\omega c} - 2I_2R + I_1R + I_3R = 0$

$-I_1R + I_2\left[2R+\frac{1}{j\omega c}\right] - I_3R = 0$ ← ②

Loop ③

$-I_3\cdot\frac{1}{j\omega c} - I_3R - I_3R + I_2R = 0$

$I_3\cdot\frac{1}{j\omega c} + 2I_3R - I_2R = 0$

$-I_2R + I_3\left[2R+\frac{1}{j\omega c}\right] = 0$ ← ③

$$\begin{vmatrix} \left[R+R_c+\frac{1}{j\omega c}\right] & -R & h_{fe}R_c \\ -R & 2R+\frac{1}{j\omega c} & -R \\ 0 & -R & 2R+\frac{1}{j\omega c} \end{vmatrix} = 0$$

$\left[R+R_c+\frac{1}{j\omega c}\right]\left\{\left(2R+\frac{1}{j\omega c}\right)\left(2R+\frac{1}{j\omega c}\right) - (R^2)\right\}$

$+R\left\{-R\left(2R+\frac{1}{j\omega c}\right) - 0\right\} + h_{fe}R_c(R^2-0) = 0$

$\left[R+R_c+\frac{1}{j\omega c}\right]\left\{4R^2+\frac{2R}{j\omega c}+\frac{2R}{j\omega c}+\frac{1}{j^2\omega^2c^2} - R^2\right\}$

$-R^2\left(2R+\frac{1}{j\omega c}\right) + h_{fe}R^2R_c = 0$

$\left[R+R_c+\frac{1}{j\omega c}\right]\left\{4R^2+\frac{4R}{j\omega c}+\frac{1}{j^2\omega^2c^2} - R^2\right\}\ldots$

$\left[R+R_c+\frac{1}{j\omega c}\right]\left\{4R^2-R^2-\frac{1}{\omega^2c^2}+\frac{4R}{j\omega c}\right]\ldots$

$\left[R+R_c+\frac{1}{j\omega c}\right]\left\{3R^2-\frac{1}{\omega^2C^2}+\frac{4R}{j\omega c}\right\}\ldots$

$\cancel{\left[R+R_c\right]\left[3R^2-\frac{1}{\omega^2C^2}\right]+\frac{1}{j\omega c}\left[3R^2-\frac{1}{\omega^2c^2}\right]}$

$\cancel{3R^3-\frac{R}{\omega^2C^2}+\frac{4R^2}{j\omega c}+3R^2R_c-\frac{R_c}{\omega^2C^2}}$

$\cancel{+\frac{4RR_c}{j\omega c}+\frac{3R^2}{j\omega c}-\frac{1}{j\omega^3C^3}+}$

$\cancel{\frac{4R}{j^2\omega^2C^2}}$ —

①

②

$$\left[R+R_c+\frac{1}{j\omega C}\right]\left\{3R^2+\frac{4R}{j\omega C}-\frac{1}{\omega^2C^2}\right\}-R^2\left\{2R+\frac{1}{j\omega C}\right\}+h_{fe}R^2R_c=0$$

$$\cancel{3R^3}+\cancel{\frac{4R^2}{j\omega C}}-\cancel{\frac{R}{\omega^2C^2}}+\cancel{3R^2R_c}+\frac{4RR_c}{j\omega C}+\cancel{\frac{R_c}{\omega^2C^2}}+\cancel{\frac{3R^2}{j\omega C}}+\cancel{\frac{4R}{j^2\omega^2C^2}}-\frac{1}{j\omega^3C^3}-\cancel{2R^3}-\cancel{\frac{R^2}{j\omega C}}+h_{fe}R^2R_c=0$$

$$R^3+\cancel{\frac{4R^2}{j\omega C}}+\frac{3R^2}{j\omega C}-\cancel{\frac{R^2}{j\omega C}}-\cancel{\frac{R}{\omega^2C^2}}-\cancel{\frac{4R}{\omega^2C^2}}+\cancel{3R^2R_c}-\cancel{\frac{R_c}{\omega^2C^2}}+\frac{4RR_c}{j\omega C}+h_{fe}R^2R_c=0$$

$$R^3+\frac{6R^2}{j\omega C}-\frac{5R}{\omega^2C^2}-\frac{R_c}{\omega^2C^2}+3R^2R_c+\frac{4RR_c}{j\omega C}+h_{fe}R^2R_c=0$$

$$R^3+3R^2R_c-\frac{1}{\omega^2C^2}\left[5R+R_c\right]+h_{fe}R^2R_c+\frac{6R^2}{j\omega C}+\frac{4RR_c}{j\omega C}-\frac{1}{j\omega^3C^3}=0$$

$$R^3+3R^2R_c-\frac{1}{\omega^2C^2}(5R+R_c)+h_{fe}R^2R_c-j\frac{6R^2}{\omega C}-j\frac{4RR_c}{\omega C}+j/\omega^3C^3=0$$

$$R^3+3R^2R_c-\frac{1}{\omega^2C^2}(5R+R_c)+h_{fe}R^2R_c-j\left\{\frac{6R^2}{\omega C}+\frac{4RR_c}{\omega C}-\frac{1}{\omega^3C^3}\right\}=0$$

fr.q of operation

$$\frac{6R^2}{\omega C}+\frac{4RR_c}{\omega C}-\frac{1}{\omega^3C^3}=0$$ multiply ωC to both side

$$6R^2+4RR_c-\frac{1}{\omega^2C^2}=0$$

$$6R^2+4RR_c=\frac{1}{\omega^2C^2} \quad \text{--------------} \quad (4)$$

$$\omega^2=\frac{1}{C^2(6R^2+4RR_c)}$$ but $K=\frac{R_c}{R}$ and $R_c=RK$

$$\omega^2=\frac{1}{C^2\{6R^2+4R(RK)\}}$$

$$\omega^2=\frac{1}{C^2(6R^2+4R^2K)}$$

$$\omega^2=\frac{1}{C^2R^2(6+4K)}$$

$$\omega=\frac{1}{RC\sqrt{6+4K}}$$ or $$f=\frac{1}{2\pi RC\sqrt{6+4K}}$$

(3)

(P.T.O)

$$R^3 + 3R^2R_c - (6R^2 + 4RR_c)(5R + R_c) + h_{fe}R^2R_c = 0$$

$$R^3 + 3R^2R_c - 30R^3 - 6R^2R_c - 20R^2R_c - 4RR_c^2 + h_{fe}R^2R_c = 0$$

$$-29R^3 - 23R^2R_c - 4RR_c^2 + h_{fe}R^2R_c = 0$$

$$h_{fe}R^2R_c = 29R^3 + 23R^2R_c + 4RR_c^2$$

$$h_{fe} = \frac{29\cancel{R^3}}{\cancel{R^2}R_c} + \frac{23\cancel{R^2R_c}}{\cancel{R^2R_c}} + \frac{4\cancel{R}R_c^{\cancel{2}}}{R^{\cancel{2}}\cancel{R_c}}$$

$$h_{fe} = \frac{29R}{R_c} + 23 + \frac{4R_c}{R} \qquad k = \frac{R_c}{R} \qquad k^{-1} = \frac{R}{R_c}$$

$$h_{fe} = \frac{29}{k} + 23 + 4k$$

$$\boxed{h_{fe}(k) = 29 + 2k + 4k^2 \qquad k = 0}$$

$$\frac{dh_{fe}}{dk} = 0 \qquad \boxed{h_{fe}(k) = 29}$$

$$-\frac{29}{k^2} + 0 + 4 = 0$$

$$-\frac{29}{k^2} = -4$$

$$k^2 = \frac{29}{4} \Rightarrow k = \sqrt{\frac{29}{4}} = 2.6925$$

(4)

$h_{fe(min)} = A \geq 44.5$

when $R_c = 0$

$$A = \frac{V_o}{V_{in}} = -\left(\frac{5}{\omega^2 R^2 C^2} - 1\right) - j\left(\frac{6}{\omega RC} - \frac{1}{\omega^3 R^3 C^3}\right)$$

at Resonant freq. imaginary part is equal to zero, then the gain becomes

$$\frac{V_o}{V_{in}} = -\left(\frac{5}{\omega^2 R^2 C^2} - 1\right) = 0 \quad \text{------ (1)}$$

freq of operation

$$\frac{6}{\omega RC} - \frac{1}{\omega^3 R^3 C^3} = 0 \quad \text{------ (2)} \Rightarrow \frac{6}{\omega RC} = \frac{1}{\omega^3 R^3 C^3}$$

$$6 = \frac{1}{\omega^2 R^2 C^2}$$

$$\omega^2 = \frac{1}{R^2 C^2 6}$$

$$\omega = \frac{1}{RC\sqrt{6}}$$

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

$$-(5 \times 6 - 1) = A$$

$$-(30 - 1) = A$$

$$-30 + 1 = A$$

$$-29 = A$$

$$A_v \geq -29$$

The gain can be obtained as

$$-\left(\frac{5}{\left(\frac{1}{\cancel{R^2C^2}6}\right)\cancel{R^2C^2}} - 1\right) = A$$

$$-\left(\frac{5}{1/6} - 1\right) = A$$

$$A_v \beta_v = 1$$

$$-29 \beta_v = 1$$

$$\beta_v = -\frac{1}{29}$$

$$f_r = \frac{1}{2\pi\sqrt{L C_T}}$$

where $C_T = \dfrac{C_1 C_2}{C_1 + C_2}$

$$\beta = \frac{C_2}{C_1} \quad \text{and} \quad A = \frac{C_1}{C_2}$$

for sustainable oscillation the loop gain

$$A\beta = 1$$

These types of oscillators give sinusoidal waveforms

Multivibrators

There are three types of multivibrators called Astable, bistable and monostable. These circuits give square waves unlike Hartley or colpitts oscillators.

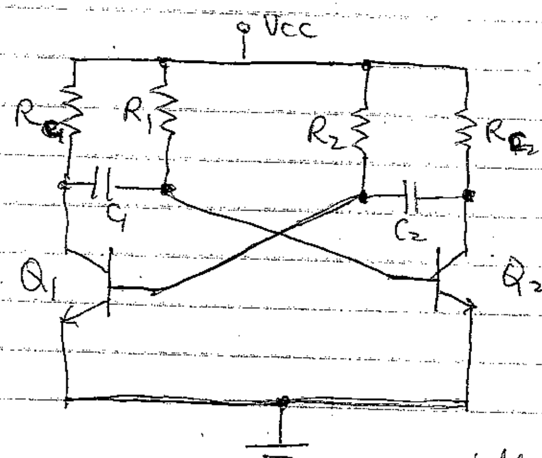
Astable: This is the circuit that oscillates between two states, HIGH and Low without any external trigger.

Bistable: This is the circuit that may be in any one of the two states indefinitely. It can only switch state to another stable state given an external trigger.

Monostable is the one which is stable in only one state. If triggered it goes into unstable state briefly depending upon the timing components before reverting to the stable state.

~~Astab~~

Astable multivibrator: Derivation of frequency of operation of the circuit



astable multivibrator circuit

If $T_1 = 0.69 R_1 C_1$ high state time constant
$T_2 = 0.69 R_2 C_2$ low state time constant
then $T = T_1 + T_2$ = time period of the circuit

For a 50% duty cycle $= \frac{T_2}{T_1 + T_2}$

$$\frac{50}{100} = \frac{T_2}{T_1 + T_2}$$

$$100T_2 = 50T_1 + 50T_2$$

$$\therefore 50T_2 = 50T_1$$

$$T_2 = T_1$$

for $T_2 = T_1$, $R_1 = R_2 = R$
$C_1 = C_2 = C$

$$\therefore T = T_1 + T_2 = 0.69RC + 0.69RC$$

$$= 1.38RC$$

But $f = \frac{1}{T} = \frac{1}{1.38RC}$

To change the frequency, keeping the same equal-mark-space, both RC time constant will be changed together at the same time. But to change both frequency and mark-space ratio, only one RC time constant network need to be changed.

Applications of multivibrators in digital circuits.
- for timing digital circuits (clock)
- for triggering digital circuits
- for synchronization of digital circuits

Example 1 - Design an astable multivibrator using transistors given the following specifications:
frequency of operation = 50 kHz
$V_{be_1} = 0.3V$ for transistor Q_1
$V_{be_2} = 0.3V$ for transistor Q_2
$V_{ce_1} = 0.3V$ for transistor Q_1
$V_{ce_2} = 0.3V$ for transistor Q_2
$\beta_1 \simeq 100$ for transistor Q_1
$\beta_2 = 100$ for transistor Q_2

(5)

(4)

$V_{cc} = 12V$

$I_{C_1} = 20mA$ for Q_1

$I_{C_2} = 20mA$ for Q_2

Let R_{C_1} and R_{C_2} be collector resistances for Q_1 and Q_2

$\therefore V_{RC_1} = V_{cc} - V_{ce_1}$

$= 12 - 0.3 = 11.7V$

$\therefore R_{C_1} = \dfrac{V_{RC_1}}{I_{C_1}} = \dfrac{11.7}{20mA} = 585\Omega$

Similarly for $R_{C_2} = \dfrac{V_{RC_2}}{I_{C_2}} = \dfrac{11.7}{20mA} = 585\Omega$

For 50% duty cycle $= \dfrac{T_2}{T_1+T_2}$

$$\frac{50}{100} = \frac{T_2}{T_1+T_2}$$

$$50T_1 + 50T_2 = 100T_2$$

$$50T_1 = 50T_2$$

$$T_1 = T_2$$

$$\therefore R_1 = R_2 = R$$

$$C_1 = C_2 = C$$

Min base current $I_{b\,min} = \dfrac{I_c}{\beta}$

but safe $I_b = 3 \times I_{b\,min} = 3 \times \dfrac{I_c}{\beta}$

$\therefore R_1 = V_{cc} - V_{be_1} / I_{b_1}$

but $I_{b_1} = 3 \times \dfrac{I_{C_1}}{\beta_1} = \dfrac{3 \times 20mA}{100} = 600\mu A$

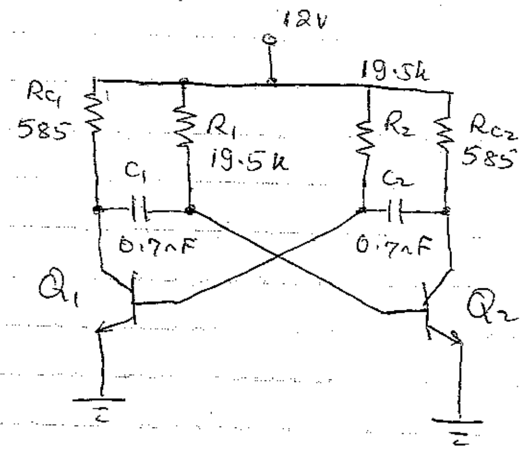
$\therefore R_1 = \dfrac{12 - 0.3}{600\mu A} = 19.5\,k\Omega$

Similarly $R_2 = 19.5k\Omega$

from $f = \dfrac{1}{1.38RC}$, $C = \dfrac{1}{1.38 \times 19.5k \times 50kHz}$

$\approx 0.7nF$

(5)



(7)

4) With negative feedback, an amplifier gives an output of 10V with an input of 0.5V. When the feedback is removed it requires 0.25V input for the same output. Calculate (i) gain without feedback (ii) feedback factor β.

5) Draw the basic negative feedback amplifier which will modify input and output impedances to match microphone and loudspeaker respectively

3/100 0·05   10%

a Draw a feedback amplifier block diagram ~~and show that~~

b Show that $A_{v_f} = \dfrac{A}{1+A\beta}$

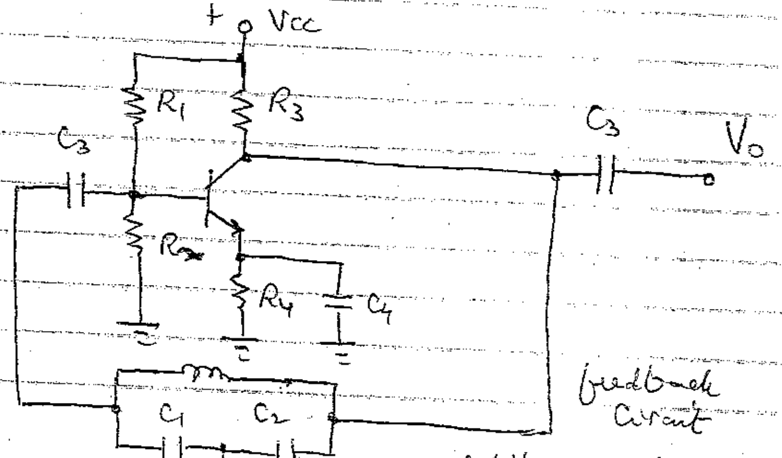
c, If the open loop gain $A = 10^5$ and the feedback factor β = 5%, find the gain with feedback.

d) If the amplifier factor of an amplifier is 10^5, what must be its feedback factor β to achieve a closed loop gain of 8.

$$\beta = \frac{L_1}{L_2} \text{ and } A = \frac{L_2}{L_1}$$

~~Col~~.

<u>Colpitts Oscillator</u>



Colpitt's oscillator